

BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2014- 2015
Comprehensive Examination

Course Code: CS F364
Course Title: Design & Analysis of Algorithms
Duration: 3 hours

III YEAR

Date: 31.05.2015
Max Marks: 40
Weightage: 40%

1. Information to be transmitted over the internet contains the following characters with their associated frequencies as shown in the following table:

Characters	a	e	l	n	o	s	t
Frequency	45	65	13	45	18	22	53

Use Huffman technique to answer the following questions:

- (a) Build the Huffman code tree for the message.
 - (b) Use the Huffman tree to find the codeword for each character.
 - (c) If the data consists of only these characters, what is the total number of bits to be transmitted? What is the percentage saving if the data is sent with 8-bit ASCII values without compression?
 - (d) Write the running time. 4M
2. A taxi hire company has one taxi at each of five depots a,b,c,d and e. A customer requires a taxi in each town, namely A,B,C,D and E. Distances (in kms) between depots (origins) and towns (Destinations) are given in the following distance matrix:

	a	b	c	d	e
A	140	110	155	170	180
B	115	100	110	140	155
C	120	90	135	150	165
D	30	30	60	60	90
E	35	15	50	60	85

How should taxis be assigned to customers so as to minimize the distance travelled?
What you mean by Hungarian method? 3M

3. What are the factors that influence the efficiency of the backtracking algorithm?
What is the running time of a program implementing the algorithm? 3M
4. Arrange 7 queens on an 7×7 chess board, so that no two queens attack each other. What is the method used to find the solution in n-queen problem by symmetry? 2M
5. Write the magic square to make the sum of any row or column or diagonal is 139. (Use 4×4 matrix and distinct positive integers) What other special features this magic square have? Write the pseudo code for the magic square. 3M

6. Apply the bottom up dynamic programming algorithm to the following instance of Knapsack problem. Capacity $W = 10$. Find out which items are in the optimal subset. What are the several algorithms available to solve knapsack problems? 3M

Item	Value	Weight
1	6	4
2	4	2
3	5	3
4	3	1
5	9	6
6	7	4

7. For the following two strings

Sequence 1 ABCBDAB

Sequence 2 BDCAB

Find the longest common subsequence. Write an algorithm to compute the length of the LCS. 3M

8. Solve the 15-puzzle using branch and bound algorithm and write the pseudo code. 2M

initial state

8	7	2	5
12	14	1	11
10	6	13	3
9	15	4	

goal state

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

9. A company makes two products (X and Y) using two machines (A and B). Each unit of X that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B. Each unit of Y that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B.

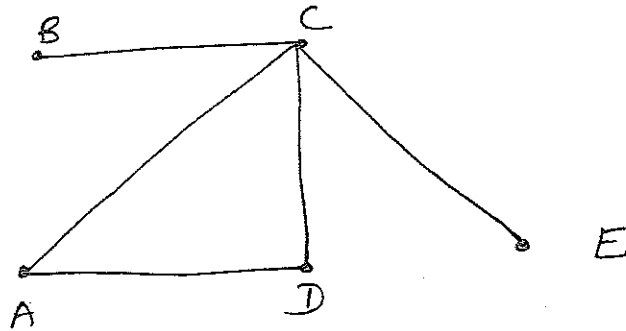
At the start of the current week there are 30 units of X and 90 units of Y in stock. Available processing time on machine A is forecast to be 40 hours and on machine B is forecast to be 35 hours.

The demand for X in the current week is forecast to be 75 units and for Y is forecast to be 95 units. Company policy is to maximize the combined sum of the units of X and the units of Y in stock at the end of the week.

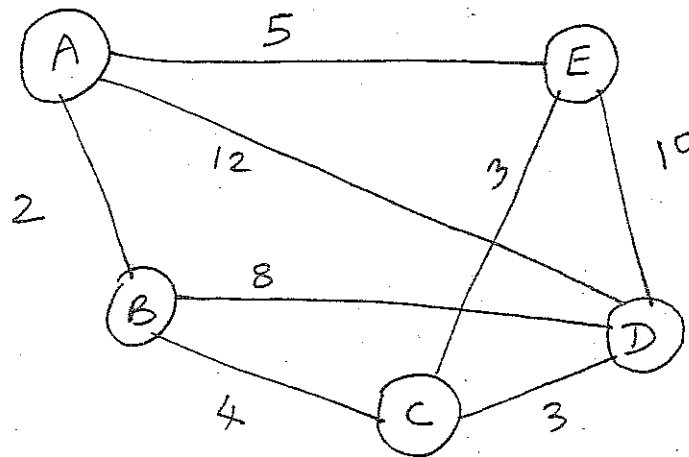
Formulate the problem of deciding how much of each product to make in the current week as a linear program. Explain about standard form in linear programming considering the parts. 3M

10. Consider the following graph G representing pairs of people (A,B,C,D and E) who are acquainted with each other:

We wish to install the minimum number of phone lines so that communication between these people is maintained. As an adviser, you need to find a spanning tree T for G . How many trees can you draw? How many edges does each tree have? Also find the minimum spanning tree. 3M



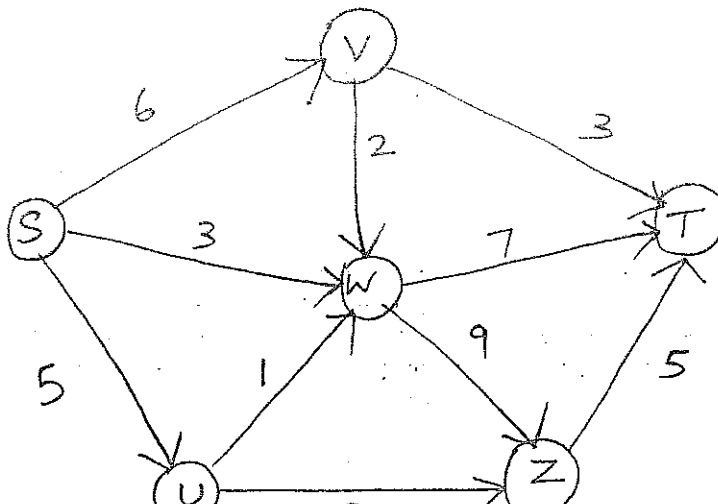
11. Consider the following set of cities. Find the minimum distance to be travelled by the salesman. 2M



12. (i) Prove Graph coloring is NP complete. 3M

(ii) Prove subset sum problem is NP hard. 3M

(iii) Explain Ford fulkerson algorithm. A company owns a factory located in city S where products are manufactured that need to be transported to the distribution centre in city T. You are given the one way roads that connect pairs of cities in the country and the maximum number of trucks that can drive along each road. What is the maximum number of trucks that the company can send to the distribution centre? 3M



BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2014- 2015

Test-II (Open Book)

Course Code: CS F364

III YEAR

Course Title: Design & Analysis of Algorithms

Duration: 50 minutes

Date: 28.04.2015

Max Marks: 20

Weightage: 20%

Note: Only Prescribed text book and hand written class notes are allowed.

1. (i) Apply the bottom up dynamic programming algorithm to the following instance of Knapsack problem. Capacity $W = 11$. Find out which items are in the optimal subset. (2 marks)

Item	Value	Weight
1	1	1
2	6	2
3	18	5
4	22	6
5	28	7

- (ii) Write the running time of Knapsack algorithm and explain how to derive it. (2 marks)

2. Try to solve the 15-puzzle using branch and bound algorithm and write the pseudo code.

initial state

2	7	12	13
	8	5	4
9	10	11	6
15	3	1	14

goal state

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

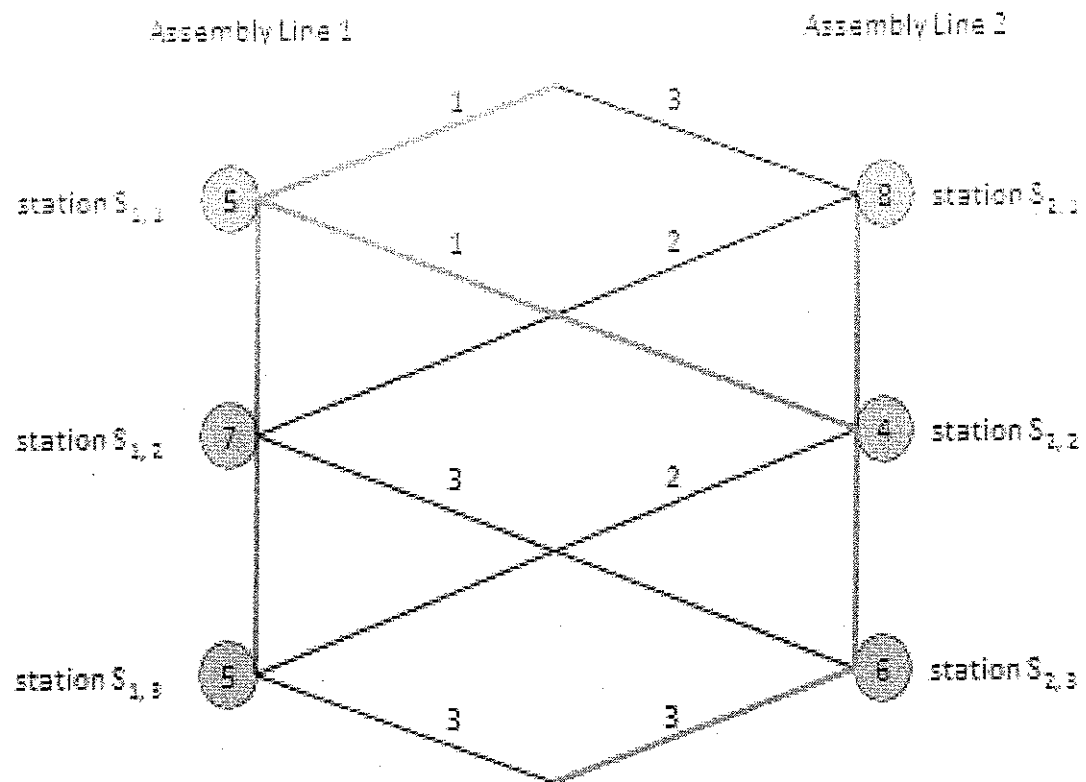
Tabulate the intermediate state and is it possible to reach the goal state from initial state. Justify your answer. (2+2 marks)

3. Prove (1.5 + 1.5 + 2 + 5 marks)

- (i) If Y is NP-complete and X is in NP and $Y < pX$, then X is NP-complete.
(ii) If X is NP-complete then X is solvable in polynomial time if and only if $P = NP$.
(iii) Knapsack problem is NP hard.

(iv) The Hamilton Cycle (HC) problem asks if there is a simple cycle that visits every node once in an undirected graph $G(V, E)$. Prove that Directed-HC (DHC) is NP-complete, where in Directed-HC we are given as input a directed graph $G'(V, E)$, and we ask if there is a simple directed cycle that visits every node once.

4. What stations should be chosen from line 1 and which from line 2 in order to minimize the total time through the factory for one car? (2 marks)



BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2014- 2015
Test-I (Closed Book)

Course Code: CS F364
Course Title: Design & Analysis of Algorithms
Duration: 50 minutes

III YEAR

Date: 02.03.2015
Max Marks: 20
Weightage: 20%

1. Show how you would use Huffman coding to encode the following string.

big brown book

- | | | |
|-------|---|------|
| (i) | What is the frequency of each character in the text? (Tabulate) | 0.5M |
| (ii) | Draw the tree. (only the final tree) | 2M |
| (iii) | Traverse tree for codes for each character. (Tabulate) | 1M |
| (iv) | Encode the file using new code words. | 0.5M |

2. For the following two strings

Sequence 1 RACECAR
Sequence 2 CREAM

- | | | |
|-------|--|----|
| (i) | Find the longest common subsequence. | 1M |
| (ii) | Construct the LCS length table. | 1M |
| (iii) | Back track the table with back pointers. | 1M |
| (iv) | Write the running time. | 1M |

3. A department manager has four jobs that she wishes to assign to four workers. the estimated costs assigning a particular worker to a particular job are shown in the table below. The assignment is made so that the total cost is minimum. Write the assignment and the cost. 4M

	1	2	3	4
Ann	80	40	50	46
Bob	40	70	20	25
Chris	30	10	20	30
Dev	35	20	25	30

4. Pack the following numbers into five bins such that the sum of the numbers in each bin is less than or equal to 150. 2M

42	63	67	57	93	90	38	36	45	42
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5. Given the adjacency matrix

A	B	C	D	E	F
0	1	0	0	1	1
1	0	1	0	1	0
0	1	0	1	0	0
0	0	1	0	0	0
1	1	0	0	0	1
1	0	0	0	1	0

- (i) Construct the graph. 1M
(ii) Find the chromatic number of the graph. 1M
(iii) Fill up the table with color names/codes like Red - 1, Green -2. 1M

Vertex	A	B	C	D	E	F
Color						

6. Arrange 8 queens on an 8 * 8 chess board, so that no two queens attack each other. 2M
7. Determine whether the given graph has a Hamiltonian circuit. 1M

BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2014- 2015
Quiz-II (Closed Book)

A

Course Code: CS F364
Course Title: Design & Analysis of Algorithms
Duration: 20 minutes

Date: 11.05.2015
Max Marks: 20
Weightage: 10%

Name: _____ ID No: _____

1. Five coach drivers, Mihi, Pat, Robert, Sarah and Tony, have to be assigned to drive five coaches for the following school trips:

- Indian Senior School is going to the Lake District.
- Our Own Junior School is going to the seaside.
- Kindergarten Junior School is going to a concert.
- Winchester Senior School is going to the museum(two coaches needed).

Mihi and Sarah would like to drive senior school children. Robert and Pat would like to go on the seaside trip. Pat and Tony would like to attend the concert. Robert and Pat would like to visit the museum. Pat and Tony would like to visit the Lake District.

The driver manager wishes to assign each driver to a trip they would like to do.

a. Draw a bipartite graph to show the trips that the drivers would like to do.

Initially Mihi and Pat are assigned to Winchester Senior School, Robert is assigned to Our Own Junior School and Tony is assigned to Indian Senior School. 2M

b. Starting from this matching use the maximum matching algorithm to find a complete matching . You must indicate clearly how the algorithm has been applied in this case. State your alternating path and complete matching. 2M

c. In any graph, if there is a perfect matching, then both the matching number and the edge cover number are _____. 1M

d. A maximum matching is also known as _____ is a matching that contains the largest possible number of _____. 1M

2. The diagram shows roads connecting 4 towns A, B, C and D. The numbers on the edges represent the lengths of the roads, in kilometers between adjacent towns. Ann lives at town A and is to visit each of the other three towns before returning to town A. She wishes to find a route that will minimize her travelling distance.

- a. Complete figure to show the shortest distances in kilometers between all pairs of towns. 1M

	A	B	C	D
A				
B				
C				
D				

- b. Find an upper bound for the minimum length of a tour of this network that starts and finishes at A. 1M
- c. Find the actual route that Ann would take in order to achieve a tour of the same length as that found in part b. 1M
- d. _____ is an application of Travelling Salesman problem. 1M

BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2014- 2015

Quiz-I (Closed Book)

A

Course Code: CS F364

III YEAR

Course Title: Design & Analysis of Algorithms

Duration: 20 minutes

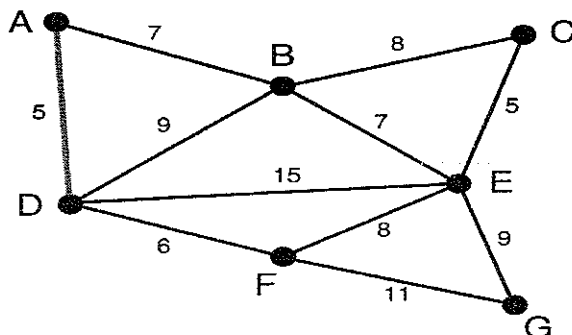
Date: 23.03.2015

Max Marks: 20

Weightage: 10%

Name: _____ ID No: _____

1. For the graph given below, use Kruskal's algorithm:



- (i) Draw the spanning tree. 0.5M
- (ii) _____ is the minimal weight. 0.5M
- (iii) Kruskal's algorithm is _____ type of algorithm. 0.5M
- (iv) What is the proof of correctness for Kruskal's algorithm. 1.5 M
- (v) Why do we need minimum spanning tree? 1M

2. A store wants to liquidate 200 of its shirts and 100 pairs of pants from last season. They have decided to put together two offers, A and B. Offer A is a package of one shirt and a pair of pants which will sell for \$30. Offer B is a package of three shirts and a pair of pants, which will sell for \$50. The store does not want to sell less than 20 packages of Offer A and less than 10 of Offer B. How many packages of each do they have to sell to maximize the money generated from the promotion? 1.5M

3. Write the magic square to make the sum -30. (Use positive and negative integers) 1.5M

4. Give one example where greedy algorithm has failed. 1.5M

5. Use Ford Fulkerson method to find the maximum flow of the network given below: When is the flow optimal? 1.5M

